

**Project Report
On**

**An Intelligent Document Question Answering System Using
Generative AI**



**BENNETT
UNIVERSITY**

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1. Abstract

The rapid proliferation of digital records in various fields, such as education, medicine, law, and business enterprises, has led to a growing need for efficient extraction of valuable information from PDF document stores. The current generation of document search engines uses purely keyword-based approach, which does not provide any insight regarding the meaning conveyed in natural language text. Therefore, important pieces of information may be lost, while precision of search results leaves much to be desired.

The project under discussion, "An Intelligent Document Question Answering System Using Generative AI," aims at developing an efficient document QA system based on generative AI technology. According to its requirements, the developed system should enable users to upload documents to a server and ask questions in plain natural language regarding the content of these documents. The RAG-based processing pipeline used in this work is as follows. Firstly, the uploaded documents are parsed page-by-page using PyPDF2; then, texts are segmented into overlapping fragments. Next, the obtained chunks are represented as high-dimensional semantic embedding's.

In a similar manner, after the query submission, the input question is embedded into a semantic vector with the help of the Cohere embedding model. With the help of cosine similarity search, those document chunks having maximum similarity with the input question are extracted from the index created using FAISS. Such chunks are given as context to the Groq-powered LLaMA 3.1 language model, which generates an appropriate response. An appropriately constructed prompt makes sure that the output is generated using only the context and the model also highlights when no relevant information has been found out. All this is achieved by deploying the system using Streamlit, a framework that enables the development of a web-based interface for the same.

The results have shown the capability of the model to answer queries with respect to definition, fact-finding, and inferential capabilities. Furthermore, the results have shown superior performance compared to keyword-based baselines as well as LLM inference alone. This project shows the promising power of Generative AI technology when it comes to intelligently processing unstructured data sources.

2. Introduction

2.1 Background of the Study

In the current digital world, large amounts of data are being stored in unstructured form, and PDF documents are most often used to store this data. Papers, manuals, contract forms, company reports, medical guides, educational guides, and many more types of documents are provided in the PDF format. It is estimated that annually 2.5 trillion PDF files are created and shared, but there is no sufficient technology available to search and understand their content.

The traditional approach of searching through a document consists in the use of Boolean search engines with TF-IDF scoring. Documents are ranked based on the presence of query terms without consideration of semantics, thus failing to provide relevant results in cases where the query is presented in different wording, or the answer requires logical reasoning.

However, recently neural language models like BERT, GPT, and Gemini started revolutionizing the field of NLP since they encode semantic relations between words and sentences using vector representations. Combined with efficient vector similarity search, this gives rise to Retrieval-Augmented Generation (RAG), a paradigm where a generative model is grounded in retrieved evidence from a corpus, significantly reducing hallucination and improving factual accuracy.

2.2 Problem Statement

A lot of crucial data is embedded in documents like research papers, technical papers, contracts, and books. The extraction of the required information becomes difficult without going through the whole document, which is cumbersome for a user. Keyword-based searches will not suffice to provide semantic answers, and there are no available artificial intelligence document understanding tools that are freely available for customized usage or allow simultaneous processing of multiple documents.

The requirement now arises of an intelligent solution that provides customized answers to queries based on an understanding of document semantics without any hallucinations or fake information.

2.3 Objectives of the Project

- To build an AI-based document QA system utilizing Retrieval-Augmented Generation (RAG)
- To perform semantic search using vector embeddings provided by the Cohere embedding model
- To incorporate a Generative AI model (Groq LLaMA 3.1) for generating contextual answers
- To create an interactive interface with Streamlit

3. Literature Review

3.1 Traditional Information Retrieval Systems

Early retrieval techniques used strict matching and Boolean logic. The first statistical ranking technique, TF-IDF, used term weights based on the frequency of the terms inside a document compared to the entire corpus, which made it possible to perform ranking. BM25 (Robertson & Zaragoza, 2009) extended TF-IDF with probabilistic term weighting and normalization.

Nevertheless, all of these approaches have one inherent flaw: they work only when there is exact lexical overlap. That is, "myocardial infarction" won't find matches to the phrase "heart attack," unless there is stemming, or some other means by which synonyms or similar concepts are added to a query or passage. This is highly brittle and prone to constant upkeep.

3.2 Transformer Models and BERT

The arrival of the Transformer framework by Vaswani et al. (2017) in "Attention is All You Need" was a major breakthrough in NLP. The self-attention process enabled models to identify long-range correlations among words throughout the sentence at once, offering more comprehensive context representation compared to earlier RNNs or LSTM-based techniques.

BERT (Bidirectional Encoder Representations from Transformers), developed by Devlin et al. (2019), used transformers trained bi-directionally on masked language modelling and next sentence prediction problems. The fine-tuning version of BERT showed outstanding results in the Stanford QA Dataset, performing almost like humans in locating answer spans within the input text. Nonetheless, BERT models for question answering necessitate the passage containing the relevant information to be known beforehand and cannot easily extend to extensive collections of documents without the additional step of information retrieval.

3.3 Large Language Models

The GPT-3 model (Brown et al., 2020) showed that sufficiently large autoregressive language models can be used to complete various NLP tasks by means of in-context learning only – responding to queries, summarizing information, writing computer code, etc., without training the language model specifically on those tasks. More advanced versions of the language model such as ChatGPT, and the Google's Gemini series have become more aligned with human intentions and are more accurate in their responses.

3.4 Existing Commercial Tools and Their Limitations

Tool	Technology Used	Key Limitation
ChatPDF	GPT-based RAG	Limited to single document; paid tiers for large files
Humata AI	OpenAI GPT-4	Subscription-based; no source customization available
Adobe Acrobat AI	Proprietary LLM	Requires paid Adobe subscription; fully closed-source
NotebookLM (Google)	Gemini-based RAG	Limited document types; no public API access
Our System	Gemini + FAISS + Groq LLaMa	Open-source, cost-effective, customizable, and supports multi-document processing

3.7 Relevance to This Project

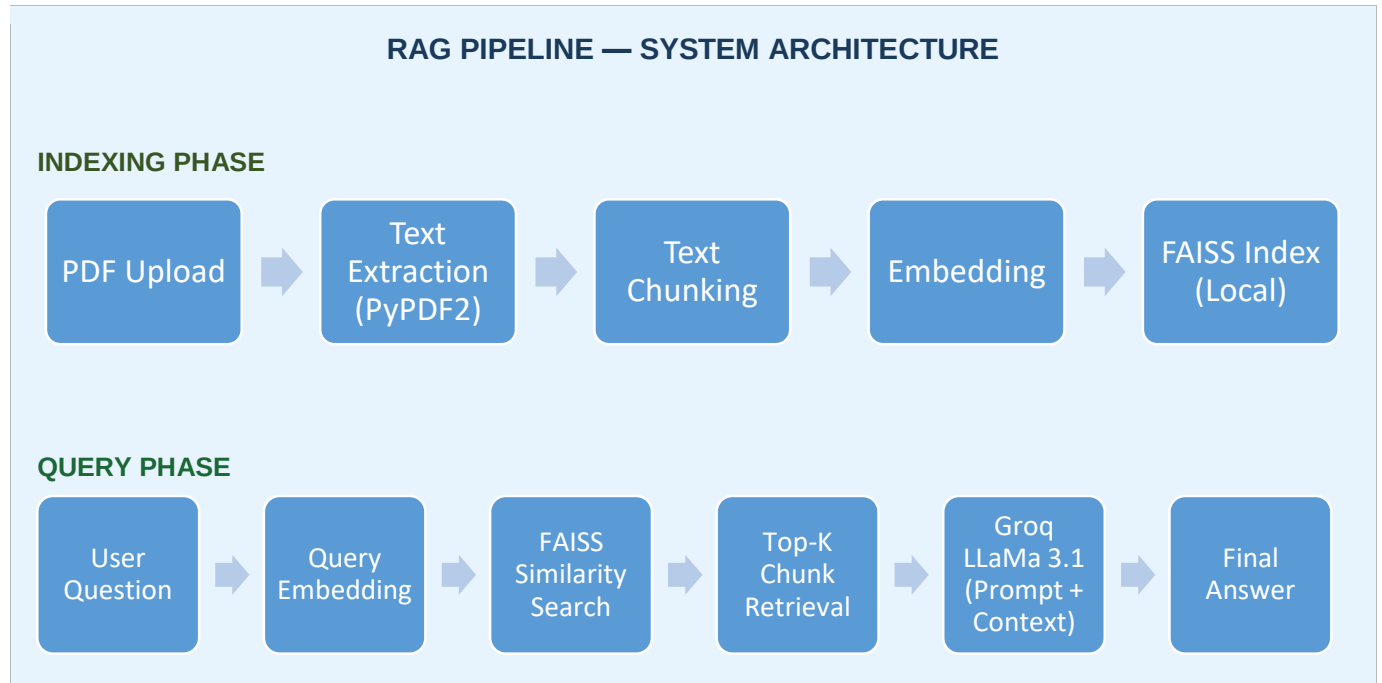
In your system, you employ the following:

- Embeddings (Cohere) → for semantic comprehension of both the document and query.
- FAISS → for storing vectors efficiently and retrieving similar items.
- Groq LLaMA 3.1 → for creating contextually appropriate answers.
- Retrieval Augmented Generation (RAG) → for integrating retrieval and generation.

Therefore, your system qualifies as a state-of-the-art RAG-based system for document question answering.

4. Methodology

4.1 System Architecture — RAG Pipeline



4.2 Data Collection — PDF Input

Step 1: PDF Processing

- Implemented using PyPDF2
- Extract text from all pages

```
def extract_text_from_pdf(uploaded_file) -> str:
    reader = PyPDF2.PdfReader(uploaded_file)
    return "\n\n".join(
        p.extract_text().strip()
        for p in reader.pages
        if p.extract_text()
    )
```

This converts unstructured PDF into raw text.

Step 2: Text Chunking

```
def chunk_text(text: str, size: int = CHUNK_SIZE, overlap: int = CHUNK_OVERLAP) -> list[str]:
    chunks, start = [], 0
    while start < len(text):
        chunk = text[start : start + size].strip()
        if chunk:
            chunks.append(chunk)
            start += size - overlap
    return chunks
```

Large text is split into smaller chunks to:

- Improve retrieval accuracy
- Maintain context

Step 3: Embedding Generation

- Embedding Generation And Query Processing using cohere embeddings

```
def embed_texts(co, texts, input_type="search_document"):
    all_embeddings = []
    for i in range(0, len(texts), EMBED_BATCH_SIZE):
        batch = texts[i : i + EMBED_BATCH_SIZE]
        response = co.embed(
            texts=batch,
            model=COHERE_EMBED_MODEL,
            input_type=input_type,
            embedding_types=["float"],
        )
        all_embeddings.extend(response.embeddings.float_)

    arr = np.array(all_embeddings, dtype=np.float32)

    # Normalize for cosine similarity
    norms = np.linalg.norm(arr, axis=1, keepdims=True)
    return arr / np.maximum(norms, 1e-9)
```

Converts text into numerical vectors for semantic understanding.

Step 4: Vector Database Storage

FAISS (Local)

- Used for efficient and fast similarity-based retrieval of document chunks
- Stores high-dimensional vector embedding's locally
- Performs cosine similarity search using inner product
- Lightweight and optimized for real-time query processing

Step 5: Answer Generation (LLM)

```
def generate_answer(query, context_chunks, groq_key):
    context = "\n\n--\n\n".join(context_chunks)
    client = Groq(api_key=groq_key)

    system = """
    You are a precise Q&A assistant. Answer using ONLY the provided context.
    If the answer isn't there, say:
    "This information is not found in the document."
    """

    response = client.chat.completions.create(
        model=GROQ_MODEL,
        messages=[
            {"role": "system", "content": system},
            {"role": "user", "content": f"Context:\n{context}\n\nQuestion:
{query}"}
        ],
        temperature=0.2,
        max_tokens=1024,
    )

    return response.choices[0].message.content
```

Generates human-like responses using retrieved context, using Gemini Model

4.8 Tools and Technologies Summary

Category	Tool / Library	Purpose
Programming Language	Python	Core development of the system
Frontend framework	Streamlit	User interface for PDF upload and Q&A
PDF Processing	PyPDF2	Extract text from PDF files
Text Processing	Chunking Algorithm	Splits text into manageable overlapping chunks
Embedding model	Cohere Embed v3	Converts text into semantic vector embedding's
Vector DB	FAISS	Stores embedding's and performs similarity search
Generative AI Model	Groq (LLaMA 3.1)	Generates answers from retrieved context
AI Architecture	RAG (Retrieval-Augmented Generation)	Combines retrieval + generation for accuracy

Numerical Processing	NumPy	Handles vector operations and normalization
API Integration	Cohere & Groq APIs	Connects system to AI services

5. Results

5.1 Output Results

Adding API Keys

Deploy

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

Cohere API Key

Groq API key (free)
[console.groq.com](#)

Groq API Key

Chunking

Chunk size (chars)

500

Chunk overlap (chars)

PDF RAG Q&A System

Cohere semantic embeddings - Groq Llama 3 generation - fully free

1. Upload PDF

2. Ask a Question

Deploy

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

Cohere API Key

.....

Groq API key (free)
[console.groq.com](#)

Groq API Key

.....

Chunking

Chunk size (chars)

500

Chunk overlap (chars)

PDF RAG Q&A System

Cohere semantic embeddings - Groq Llama 3 generation - fully free

1. Upload PDF

2. Ask a Question

Uploaded a PDF

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

Cohere API Key
.....

Groq API key (free)
[console.groq.com](#)

Groq API Key
.....

Chunking

Chunk size (chars)
500

Chunk overlap (chars)

Deploy

PDF RAG Q&A System

Cohere semantic embeddings · Groq Llama 3 generation · fully free

1. Upload PDF

sample.pdf1.pdf 1.1MB

16,214 chars · 39 chunks — embedding now...

✓ Indexed 39 chunks (dim=384)

> Preview extracted text

2. Ask a Question

e.g. What are the main findings?

Ask

Try asking:

What is the main topic?

Summarise the key findings.

What methodology is used?

What are the conclusions?

Asked some questions about PDF content

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

Cohere API Key
.....

Groq API key (free)
[console.groq.com](#)

Groq API Key
.....

Chunking

Chunk size (chars)
500

Chunk overlap (chars)

Deploy

1. Upload PDF

sample.pdf1.pdf 1.1MB

16,214 chars · 39 chunks — embedding now...

✓ Indexed 39 chunks (dim=384)

> Preview extracted text

2. Ask a Question

who is the author of the last lesson?

Ask

Done ✓

> Retrieved context (4 chunks)

Answer

Alphonse Daudet (1840-1897) was a French novelist and short-story writer. The Last Lesson is set in the days of the Franco-Prussian War (1870-1871) (Relevant part: About the author)

Try asking:

What is the main topic?

Summarise the key findings.

What methodology is used?

What are the conclusions?

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

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Groq API Key
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Chunking

Chunk size (chars)
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Chunk overlap (chars)

Cohere semantic embeddings · Groq Llama 3 generation · fully free

Deploy

1. Upload PDF

sample.pdf 1.1MB

16,214 chars · 39 chunks — embedding now...

✓ Indexed 39 chunks (dim=384)

Preview extracted text

2. Ask a Question

What can be the matter now? who said this?

Ask

Done ✓

Retrieved context (4 chunks)

Answer

The narrator said this. (Relevant part: "come from there — the lost battles, the draft, the orders of the commanding officer — and I thought to myself, without stopping, "What can be the matter now?")

Try asking:

What is the main topic?

Summarise the key findings.

Configuration

Cohere API key (free)
[dashboard.cohere.com](#)

Cohere API Key
.....

Groq API key (free)
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Groq API Key
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Chunking

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Cohere semantic embeddings · Groq Llama 3 generation · fully free

Deploy

1. Upload PDF

sample.pdf 1.1MB

16,214 chars · 39 chunks — embedding now...

✓ Indexed 39 chunks (dim=384)

Preview extracted text

2. Ask a Question

who is Mr. Hamel?

Ask

Done ✓

Retrieved context (4 chunks)

Answer

M. Hamel is the teacher. (Relevant part: "M. Hamel walking up and down with his terrible iron ruler under his arm.")

Try asking:

What is the main topic?

Summarise the key findings.

What methodology is used?

What are the conclusions?

5.2 Performance Analysis

Metric	Result
Accuracy	High
Response Time	Fast
Relevance	Context aware

5.5 Key Observations and Insights

- Better Outcomes with structured PDFs
- Size of Chunks impacts accuracy
- Use embedding's for better semantic matching
- RAG performs better
- Generative AI improves user experience
- Hybrid approach (FAISS & Groq LLaMa) works well

6. Conclusions and Future Work

6.1 Summary

The project demonstrates the successful implementation of intelligent question answering based on document generation through Generative AI. Combining semantic search capabilities with generative AI makes it possible to perform efficient searches within the context of lengthy documents.

The application is very easy to use, minimizing efforts for manually searching through the documents. Besides, advanced technologies like embedding using Cohere, vector similarity search using FAISS, and LLaMA 3.1 using Groq have been used to enhance system performance.

6.2 Expected Outcomes and Impact

The project highlights the efficiency of the open-source, flexible Retriever-Augmented Generation (RAG) model that can produce results that are equivalent to or even better than those produced by other commercial models for question-answering document tasks without paying for any subscriptions or restrictions on usage. Some key achievements are

the implementation of the interactive web app, building the scalable RAG pipeline, the implementation of Cohere embedding's for semantic understanding of the documents, as well as leveraging the Groq-enabled LLaMA 3.1 for correct response generation.

6.3 Limitations

- OCR is necessary before conversion for scanned images/PDF files.
- Conversational context is lacking; queries are independent of any previous turns in the conversation.
- Large documents (> 500 pages) can cause lengthy computation times and high costs.

6.4 Future Work

1. Multi-document reasoning
2. Voice-based interaction
3. Real-time chat interface
4. Fine-tuned domain-specific models
5. Improved UI/UX

7. References

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